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MAT111
(Pure Mathematics)
MOCK TEST 3 [2021-22]

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BRAINSTORM GROUP (BSG) MAT111 TEST 3 (2021/22)

1. $2^{y-x} = 16$ *and* $4^{y+x} = 8$, find the value of x + y

   A. 3 B. $\frac{3}{2}$

   C. -3 D. $\frac{3}{2}$

2. *Find x in* $log_{10}(x-9) + log_{10}x = 1$

   A. 1 B. -1

   C. 2 D. -10

3. $log_{10}(x-14) + log_{10}(x-5) = 1$

   A. 4 *or* 15 B. 4 *or* -15

   C. -4 *or* 15 D. -4 *or* -15

4. $3 \times 5^x = {}^{3}/_{125}$, *what is the value of x?*

   A. $-\frac{1}{3}$ B. $\frac{1}{3}$

   C. 3 D. -3

5. $32^{2/5} \times 64^{1/18}$

   A. $\sqrt[3]{256}$ B. $\sqrt[3]{128}$

   C. $\sqrt[3]{64}$ D. $\sqrt[3]{16}$

6. *Find M given* $logM = \frac{1}{4}log81$

   A. $-\frac{1}{3}$ B. $\frac{1}{3}$

   C. 3 D. -3

7. $log_x(8x-3) - log_x4 = 2$

   A. $-\frac{1}{2}$ *or* $-\frac{3}{2}$ B. $-\frac{1}{2}$ *or* $\frac{3}{2}$

   C. $-\frac{1}{2}$ *or* $\frac{3}{2}$ D. $\frac{1}{2}$ *or* $\frac{3}{2}$

8. $log_3 27 - log_3 9 + log_3 81$

   A. 5^3 B. 3^5

   C. 5 D. -5

9. $log_a 36 + \frac{1}{2}log_a 256 - 2log_a 48$

   A. $-log_a 4$ B. $log_a 16$

   C. $log_a 8$ D. $-4log_a 2$

10. *Find the value for x in* $4^x - 7 \times 2^x + 12 = 0$

    A. 2 B. 3

    C. 4 D. 5

11. Given $2 - 3i = \sqrt{a + bi}$. Find a and b.

    A. -5 and -12 B. 5 and -12

    C. -5 and 12 D. 5 and 12

12. If $\frac{2+j}{1-j} = j(x + yj)$. Find y and x.

    A. $\frac{1}{2}, \frac{3}{2}$ B. $\frac{3}{2}, \frac{1}{2}$

    C. $\frac{1}{2}, -\frac{3}{2}$ D. $-\frac{1}{2}, \frac{3}{2}$

13. Determine the modulus, argument of $2 + 4i$.

    A. $2\sqrt{5}$, $63.43°$

    B. $2\sqrt{10}$, $63.43°$

    C. $2\sqrt{5}$, $26.57°$

    D. $2\sqrt{10}$, $26.57°$

14. Simplify $(i-1)^4$.

    A. $-4i$ B. -4

    C. $4i$ D. 4

15. If $Z_1 = 1 - i$ *and* $Z_2 = 2i - 1$, determine $\frac{Z_1 Z_2}{Z_1 + Z_2}$ in the form $a + ib$

    A. $3 - i$ B. $-3 + i$

    C. $3 + i$ D. $-3 - i$

16. All inhabitants of a certain village near the Cameroon border speak either French or English or both. If 64% speak French and 58% speak English. What is percentage speak both languages?

    A. 18 B. 22

    C. 23 D. 25

17. Given that $X = \{x : 3 < x < 6\}$ *and* $Y = \{x : 4 \leq x \leq 8\}$. *Find* $X \cap Y$.

    A. {4} B. {4, 5}

    C. {5, 6} D. {4, 5, 6, 7, 8}

18. In a class of 50 students, 25 like oranges, 22 like bananas and 4 like both oranges and bananas. How many like neither oranges nor banana?

    A. 1 B. 7

    C. 11 D. 12

19. In a class, 16 students offer geography, 18 offer chemistry, 5 offer both and 3 offer neither. How many students are in the class?

    A. 27 B. 33

    C. 32 D. 42

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20. The sum of the first five terms of an A.P. is 30 and the sum of the next four terms is 60, find the sum of the first twenty terms

A. 220 B. 320
C. 360 D. 420

21. The second term of a G.P. is 49 and the fourth term is 2401. Find the common ratio.

A. -7 B. -14
C. 14 D. 4

22. $\frac{16}{9}, x, 1, y$ are G.P. Find the product of x and y.

A. 1 B. $\frac{1}{9}$
C. $-\frac{2}{3}$ D. 3

23. The nth term of A.P. $-4, -2, 0 \ldots$ is

A. $2n-2$ B. $2n+4$
C. $2n-4$ D. $2n-6$

24. The first winner of a lottery gets €2,000 while the subsequent winners gets €50 less than the preceding winner. The winner in the 20th position gets

A. €2050 B. €1050
C. €5010 D. €2500

25. Given an A.P. $a, 4, b, 20$ Find $a+b$

A. -4 B. 4
C. 8 D. 12

26. The first term of a G.P. is 10 and the sum to infinity is 20. Find the common difference

A. $\frac{1}{3}$ B. $\frac{1}{2}$
C. $-\frac{1}{3}$ D. $\frac{1}{4}$

27. Evaluate $\sum_{r=1}^{4} r(r^2+1)$.

A. 100 B. 110
C. 120 D. 130

28. Obtain the sum of positive even integers less than 100.

A. 4950 B. 5050
C. 5500 D. 2450

29. Express $\frac{(n+2)!}{n!}$

A. $(n+1)(n+2)$ B. $\frac{1}{2}(n+1)(n+2)$
C. $(n-1)(n-2)$ D. $n(n-1)(n-2)$

30. Express $cos^2\emptyset$ terms of multiple angles

A. $2^{-1}(cos2\emptyset - 1)$ B. $2(cos2\emptyset - 1)$
C. $2^{-1}(cos2\emptyset + 1)$ D. $2(cos2\emptyset + 1)$

31. Obtain the first three terms of the expansion$(1+x)^{-m}$.

A. $1 - mx - \frac{1}{2}m(m+1)x^2$
B. $1 + mx + \frac{1}{2}m(m-1)x^2$
C. $1 - mx + \frac{1}{2}m(m+1)x^2$
D. $1 + mx + \frac{1}{2}m(m-1)x^2$

32. Obtain the fourth term of the expansion $(2+3x)^{11}$

A. $1140480x^2$ B. $1104480x^4$
C. $1140480x^3$ D. $1140480x^5$

33. Find the first three terms of the expansion $(1+x^2)^{-2}$

A. $1 - 2x^2 + 3x^4$ B. $1 - 2x^2 - 3x^4$
C. $1 + 2x^2 - 3x^4$ D. $1 + 2x^2 + 3x^4$

34. If $(1-2x)^6 = 1 + Px + Qx^2 \ldots$ find the values of the constants P and Q.

A. 60 *and* -12 B. -60 *and* 12
C. 12 *and* -60 D. -12 *and* 60

35. What is the coefficient of x^2 in $(1+x)^{-2}$.

A. 2 B. -2
C. 3 D. -3

36. Find the term independent of x in the expansion of $(x^2 - \frac{2}{x})^6$.

A. -16 B. 16
C. -240 D. 240

37. What is the coefficient of x^3 in $(2+3x)^4$.

A. 72 B. 108
C. 180 D. 216

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38. Find the term independent of x in the expansion of $(x - \frac{3}{x})^8$.

A. 5760 B. 5670
C. −5670 D. −5760

39. Calculate the coefficient of x^3y^2 in $(2x - 3y)^5$.

A. 10 B. −108
C. 720 D. −720

40. What is the coefficient of x^2 in $(1 + x)^{1/3}$.

A. $-\frac{1}{3}$ B. $\frac{1}{3}$
C $-\frac{1}{9}$ D. $\frac{1}{9}$

41. If the roots of the equation $2x^2 + 7x + 3 = 0$ are α *and* β, find the equation having roots $(2\alpha + \beta)$ *and* $(2\beta + \alpha)$

A. $2x^2 - 21x + 52 = 0$
B. $2x^2 + 21x - 52 = 0$
C. $2x^2 - 21x - 52 = 0$
D. $2x^2 + 21x + 52 = 0$

42. Find the possible values of a for which the quadratic equation $ax^2 + (a + 1)x + a = 0$ has equal roots

A. $\frac{1}{3}$ B. $-\frac{2}{3}$
C. 1 D. −1

43. If α *and* β are the roots of the equation $2x^2 - 6x + k = 0$, find the value of k if $\frac{1}{\alpha} + \frac{1}{\beta} = -3$

A. −1 B. 1
C. −2 D. 2

44. If the roots of the equation $4x^2 - 6x + 1 = 0$ are α *and* β, find the equation the roots of which are $\left(\alpha + \frac{1}{\beta}\right)$ *and* $\left(\beta + \frac{1}{\alpha}\right)$

A. $4x^2 - 30x - 25 = 0$
B. $4x^2 - 30x + 25 = 0$
C. $4x^2 + 30x - 25 = 0$
D. $4x^2 + 30x + 25 = 0$

45. If $\frac{x-5}{(x^2-4)(x^2-9)} = \frac{A}{x-2} + \frac{B}{x+2} + \frac{C}{x-3} + \frac{D}{x+3}$, find the value (A, B).

A. $\left(\frac{3}{20}, -\frac{7}{20}\right)$ B. $\left(-\frac{7}{20}, -\frac{1}{15}\right)$
C. $\left(-\frac{7}{20}, \frac{3}{20}\right)$ D. $\left(-\frac{7}{20}, \frac{4}{15}\right)$

46. If $\frac{x^2}{(x+1)^3} = \frac{A}{(x+1)} + \frac{B}{(x+1)^2} + \frac{C}{(x+1)^3}$, find (B, C).

A. (−2, 1) B. (1, −2)
C. (−1, 2) D. (2, −1)

47. Given that $\frac{3x^2-3x-2}{(x-1)(x+1)} = A + \frac{Bx+C}{(x-1)(x+1)}$, what is the value of A, B and C.

A. (3, 3, −2) B. (2, 3, 1)
C. (3, −3, − 1) D. (3, −3, 1)

48. The partial fraction of $\frac{6x-10}{x^2-2x-3}$ is

A. $\frac{2}{x-3} - \frac{4}{x+1}$ B. $\frac{4}{x-3} - \frac{2}{x+1}$
C. $\frac{2}{x-3} + \frac{4}{x+1}$ D. $\frac{4}{x-3} + \frac{2}{x+1}$

49. Given $\frac{7-2x}{(x+1)(x-2)^2} = \frac{A}{x+1} + \frac{B}{x-2} + \frac{C}{(x-2)^2}$, find A, B and C

A. −1, 1, −1 B. 1, 1, −1
C. 1, 1, 1 D. 1, −1, 1

50. Partial fraction equivalent of $\frac{1}{(ax+b)^2}$ is

A. $\frac{(Ax+B)^2}{ax+b}$
B. $\frac{Ax+B}{ax+b}$
C. $\frac{A}{(ax+b)^2} + \frac{B}{(ax+b)^3}$
D. $\frac{A}{ax+b} + \frac{B}{(ax+b)^2}$

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ANSWER									
1.	B	11.	A	21.	A	31.	C	41.	D
2.	B	12.	D	22.	A	32.	C	42.	C
3.	A	13.	A	23.	D	33.	A	43.	C
4.	D	14.	B	24.	B	34.	D	44.	B
5.	B	15.	A	25.	C	35.	C	45.	A
6.	C	16.	B	26.	B	36.	D	46.	A
7.	D	17.	B	27.	B	37.	D	47.	D
8.	C	18.	B	28.	D	38.	B	48.	C
9.	A	19.	C	29.	A	39.	C	49.	D
10.	A	20.	D	30.	C	40.	C	50.	D

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1) $2^{y-x} = 16$ and $4^{y+x} = 8$

$2^{y-x} = 2^4$ and $2^{2(y+x)} = 2^3$

equating powers

$y - x = 4$ —(i) $\quad 2(y+x) = 3$ —(ii)

Hence, from (ii): $x + y = 3/2$

B

2) $\log_{10}(x-9) + \log_{10} x = 1$

Recall: $\log_{10} 10 = 1$ and $\log A_{10} + \log B_{10} = \log AB_{10}$.

$\log_{10}[x(x-9)] = \log_{10} 10$

$x(x-9) = 10$

$x^2 - 9x - 10 = 0$

$x^2 - 10x - x - 10 = 0$

$x(x-10) + 1(x-10) = 0$

$x = -1$ or 10

B

3) $\log_{10}(x-14) + \log_{10}(x-5) = 1$

Recall: $\log_{10} 10 = 1$ and $\log_{10} A + \log_{10} B = \log_{10} AB$.

$\log_{10}[(x-14)(x-5)] = \log_{10} 10$

$(x-14)(x-5) = 10$

$x^2 - 19x + 70 - 10 = 0$

$x^2 - 19x + 60 = 0$

$x^2 - 15x - 4x + 60 = 0$

$x(x-15) - 4(x-15) = 0$

$(x-4)(x-15) = 0$

$x = 4$ or 15

A

4) $3 \times 5^x = 3/125$

$3(5^x) = 3(5^{-3})$

$5^x = 5^{-3}$

$x = -3$

D

5) $32^{2/5} \times 64^{1/18}$

$= (2^5)^{2/5} \times (2^6)^{1/18}$

$= 2^2 \times 2^{1/3}$

$= 2^{2\frac{1}{3}} = 2^{7/3}$

$= \sqrt[3]{2^7} = \sqrt[3]{128}$

B

6) $\log M = \frac{1}{4}\log 81$

$\log M = \log(81)^{1/4}$

$M = (81)^{1/4} = (3^4)^{1/4}$

$M = 3$

C

7) $\log_x(8x-3) - \log_x 4 = 2$

$\log_x\left(\frac{8x-3}{4}\right) = 2$

changing the equation in logarithm form to an equation in indices form, we have:

$x^2 = \frac{8x-3}{4}$

Hence $4x^2 - 8x - 3 = 0$

$4x^2 - 6x - 2x - 3 = 0$

$2x(2x-3) - 1(2x-3) = 0$

$(2x-1)(2x-3) = 0$

$x = +1/2$ or $+3/2$

D

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8) $\log_3 27 - \log_3 9 + \log_3 81$

$= \log_3(27 \div 9 \times 81) = \log_3 243$

$= \log_3 3^5 = 5\log_3 3 = 5$

C

9) $\log_a 36 + \frac{1}{2}\log_a 256 - 2\log_a 48$

$= \log_a 36 + \log_a 256^{1/2} - \log_a 48^2$

$= \log_a 36 + \log_a 16 - \log_a 2304$

$= \log_a\left(\frac{36 \times 16}{2304}\right) = \log_a(1/4)$

$= \log_a 4^{-1} = -\log_a 4$

A

10) $4^x - 7 \times 2^x + 12 = 0$

$2^{2x} - 7 \times 2^x + 12 = 0$

$(2^x)^2 - 7(2^x) + 12 = 0$

Let $2^x = a$

Then $a^2 - 7a + 12 = 0$

$a^2 - 4a - 3a + 12 = 0$

$a(a-4) - 3(a-4) = 0$

$(a-3)(a-4) = 0$

$a = 3$ or 4

Recall that $a = 2^x$.

Hence, when $a = 4$

$2^x = 4 = 2^2$

equating powers

$x = 2$

A

11) $2 - 3i = \sqrt{a+bi}$

squaring both sides

$(\sqrt{a+bi})^2 = (2-3i)^2$

$a + bi = (2-3i)(2-3i)$

$= 4 - 12i - 9i^2$

$= -5 - 12i$

comparing both real and imaginary parts of both sides, we have

$a = -5$ and $b = -12$

A

12) $\frac{2+j}{1-j} = j(x + yj)$

$xj + yj^2 = \frac{2+j}{1-j} \times \frac{1+j}{1+j}$

$xj - y = \frac{2 + 2j + j + j^2}{1^2 + 1^2}$

$xj - y = \frac{2 + 3j - 1}{2} = \frac{1 + 3j}{2}$

$xj - y = \frac{3}{2}j + \frac{1}{2}$

comparing the real and imaginary parts of both sides, we have:

$x = \frac{3}{2}$ and $y = -\frac{1}{2}$

D

13) $z = 2 + 4i$

Modulus of z:

$|z| = \sqrt{2^2 + 4^2} = \sqrt{20}$

$= 2\sqrt{5}$

Argument of z:

$\theta = \tan^{-1}(4/2) = \tan^{-1} 2$

$= 63.43°$

A

(14) Simplify $(i-1)^4$

$(i-1)^2 = (i-1)(i-1) = i^2 - 2i + 1$

$= -1 - 2i + 1 = -2i$

$(i-1)^4 = [(i-1)^4]^2 = (-2i)^2$

$= 4i^2 = -4$ **B**

(15) $z_1 = 1 - i$ $z_2 = 2i - 1$

$z_1 + z_2 = (1-i) + (2i-1) = i$

$z_1 z_2 = (1-i)(2i-1)$

$= 2i - 1 - 2i^2 + i = 3i + 1$

$\frac{z_1 z_2}{z_1 + z_2} = \frac{3i+1}{i} = \frac{3i+1}{i} \times \frac{i}{i}$

$= \frac{3i^2 + i}{i^2} = \frac{-3+i}{-1} = 3 - i$

A

(16) Let F represent inhabitant speaking French
Let E represent inhabitant speaking English
Then

$n(F) = 64\%$ and $n(E) = 58\%$

Also
Inhabitant speaking both language is $n(F \cap E)$
Inhabitant speaking either language is $n(F \cup E)$
Total inhabitant constitute 100%
i.e $n(U) = 100\%$

Since all inhabitant speaks either language, we have

$n(F \cup E) = n(U) = 100\%$.

Also

$n(F \cup E) = n(F) + n(E) - n(F \cap E)$

$100\% = 64\% + 58\% - n(F \cap E)$

$n(F \cap E) = 64 + 58 - 100 = 22\%$ **B**

(17) $X = \{x : 3 < x < 6\} = \{4, 5\}$

$Y = \{x : 4 \leq x \leq 8\} = \{4, 5, 6, 7, 8\}$.

Hence, $X \cap Y = \{4, 5\}$ **B**

(18) Totat number of students $n(U) = 50$
Students that like oranges, $n(O) = 25$
Students that like banana, $n(B) = 22$
Students that like both fruit, $n(O \cap B) = 4$

Hence

$n(U) = n(O) + n(B) - n(O \cap B) + n(O \cup B)'$

where $(O \cup B)$ stands for students that like nieither of the fruti

$n(O \cup B)' = n(U) - n(O) - n(B) + n(O \cap B)$

$n(O \cup B)' = 50 - 25 - 22 + 4$

$= 7$ **B**

(19) Students that offers Geography, $n(G) = 16$
Students that offers Chemistry, $n(C) = 18$
Students that offers both Subject

$n(G \cap G) = 5$

Studenats that offer none of the two Subjects

$n(G \cup G)' = 3$

Total number of students, $n(U) = ?$

$n(U) = n(G) + n(C) - n(G \cap C) + n(G \cup C)'$

$n(U) = 16 + 18 - 5 + 3$

$= 32$

Total number of students is 32 **C**

(20) $S_5 = 30$

and $S_9 = S_5 + 60 = 30 + 60 = 90$

Recall : $S_n = \frac{n}{2}[2a + (n-1)d]$

Hence

$S_5 = \frac{5}{2}[2a + (5-1)d] = \frac{5}{2}(2a+4d)$

$30 = 5(a + 2d)$

$a + 2d = 6$ —— (i)

Also

$S_9 = \frac{9}{2}[2a + (9-1)d] = \frac{9}{2}(2a+8d)$

$90 = 9(a + 4d)$

$a + 4d = 10$ —— (ii)

(ii) − (i) : $2d = 4$ $\quad d = 2$

Putting $d = 2$ in (i) :.

$a + 2(2) = 6$ $\quad a = 2$

Hence

$S_{20} = \frac{20}{2}[2(2) + (20-1)2]$

$S_{20} = 10[4 + 19(2)] = 10(4+38)$

$S_{20} = 420$ **D**

(21) Second term, $T_2 = ar = 49$

Fourth term, $T_4 = ar^3 = 2401$

by considering the general formula for nth term of a G.P to be

$T_n = ar^{n-1}$

Then $\frac{T_4}{T_2} = \frac{ar^3}{ar} = r^2 = \frac{2401}{49} = 49$

$r = \sqrt{49} = \pm 7$ **A**

(22) For a three consecutive terms in a G.P., the square of the middle term equals the product of the other two terms.

Let say, the three consecutive terms of a G.P are $a, ar, ar^2 \ldots$

Then $T_{n-1} = a \mid T_n = ar \mid T_{n+1} = ar^2$

$T_{n-1} \times T_{n+1} = a \times ar^2$

$= a^2r^2 = (ar)^2$

$= T_n^2$.

Now considering terms $16, x, 1, y \ldots$

$xy = 1^2$ $\quad xy = 1$ **A**

(23) For A.P. $-4, -2, 0 \ldots$

first term, $a = -4$

Common difference, $d = -2-(-4) = 2$

nth term $T_n = a + (n-1)d$

$T_n = -4 + (n-1)2 = -4 + 2n - 2$

$T_n = 2n - 6$ **D**

(24) first winner gets $a = 2000$

common difference, $d = -50$

Since nth term $T_n = a + (n-1)d$

The winner in the 20th position is T_{20}

$T_{20} = 2000 + (20-1)(-50)$

$= 2000 + 19(-50)$

$= 2000 - 950$

$= €1050$ **B**

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25) For three consecutive terms in an A.P., the double of the middle term equals the sum of the other two terms

Let say the three consective terms of an A.P. are $a-d,\ a,\ a+d$

Then $T_{n-1} = a-d \mid T_n = a \mid T_{n+1} = a+d$

$$T_{n-1} + T_{n+1} = (a-d) + (a+d)$$
$$= 2a$$
$$= 2T_n$$

Now considering terms $a, 4, b, 20\ldots$

$a+b = 2(4) \quad a+b = 8$ **C**

26) $a = 10 \qquad S_\infty = 20$

$$S_\infty = \frac{a}{1-r}, \qquad 20 = \frac{10}{1-r}$$

$$1-r = \frac{10}{20} = \frac{1}{2},$$

$$r = 1-\frac{1}{2} \qquad r = 1/2$$ **B**

27) $\sum_{r=1}^{4} r(r^2+1)$

r	r^2	r^2+1	$r(r^2+1)$
1	1	2	2
2	4	5	10
3	9	10	30
4	16	17	68
			110

B

28) Sum of positive integers less than 100

The series is: $2 + 4 + 6 + \ldots + 96 + 98$

first term, $a = 2$

Last term, $l = 98$

common difference $= 4 - 2$

To find the number of terms n present in the series:

$$T_n = l = a + (n-1)d$$
$$98 = 2 + (n-1)2$$
$$98 = 2 + 2n - 2 = 2n$$
$$n = 49$$

The number of terms is 49

Hence, using $S_n = \frac{n}{2}(a+l)$

$$S_{49} = \frac{49}{2}(2+98) = \frac{49}{2}(100)$$

$S_{49} = 2450$ **D**

29) $\frac{(n+2)!}{n!} = \frac{(n+2)(n+1)n!}{n!}$

$= (n+1)(n+2)$ **A**

30) From multiple angles:

$\cos(\phi + \phi) = \cos 2\phi$

$= 2\cos^2\phi - 1$

$2\cos^2\phi = \cos 2\phi + 1$

$\cos^2\phi = \frac{\cos 2\phi + 1}{2}$

$\cos^2\phi = 2^{-1}(\cos 2\phi + 1)$ **C**

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For an expansion $(a+b)^n$, term $(r+1)$ is given as:

$$\boxed{T_{r+1} = {}^nC_r\, a^{n-r}\, b^r}$$

For expansion of form $(1+a)^n$ where n is negative or fraction, use

$$(1+a)^n = 1 + \frac{n}{1}a + \frac{n(n-1)}{2}a^2 + \frac{n(n-1)(n-2)}{(2)(3)}a^3 + \dots$$

31) $(1+x)^{-m}$

$= 1 + \frac{-m}{1}x + \frac{(-m)(-m-1)}{2}x^2 \dots$

$= 1 - mx + \frac{m(m+1)}{2}x^2$ **C**

32) Fourth term, T_4 of $(2+3x)^{11}$

$T_4 = {}^{11}C_3 (2)^8 (3x)^3$ **C**

$= (165)(256)(27x^3) = 1140480x^3$

33) $(1+x^2)^{-2}$

$= 1 + \frac{-2}{1}x^2 + \frac{(-2)(-2-1)}{2}(x^2)^2 \dots$

$= 1 - 2x^2 + 3x^4 \dots$ **A**

34) $(1-2x)^6$

$= 1 + \frac{6}{1}(-2x) + \frac{(6)(5)}{2}(-2x)^2 \dots$

$= 1 - 12x + 60x^2$

$= 1 + Px + Qx^2$

comparing coefficient of x and x^2,

we have: $P = -12$ and $Q = 60$ **D**

35) The coefficient of x^2 is the third term

$(1-x)^{-2} = 1 + \frac{-2}{1}x + \frac{(-2)(-3)}{2}x^2$

$= 1 - 2x + 3x^2 \dots$

The coefficient of x^2 is 3 **C**

36) For expansion $(x^2 - \frac{2}{x})^6$,

$T_{r+1} = {}^6C_r (x^2)^{6-r} (-\frac{2}{x})^r$

$= {}^6C_r\, x^{12-2r} \cdot x^{-r} (-2)^r$

$= {}^6C_r\, x^{12-3r} (-2)^r$

In order to get an independent term of x,

$x^{12-3r} = 1 = x^0$

then $12 - 3r = 0$ $\quad r = 4$

Hence, the term independent of x is

$T_{4+1} = {}^6C_4\, x^{12-3(4)} (-2)^4$

$T_5 = (15)\, x^0\, (16)$

$= 240$ **D**

37) For expansion $(2+3x)^4$

$T_{r+1} = {}^4C_r (2)^{4-r} (3x)^r$

for coefficient of x^3, $r = 3$

Then

$T_{3+1} = {}^4C_3 (2)^{4-3} (3x)^3$

$T_4 = (4)(2)(27x^3) = 216x^3$

The coefficient of x^3 is 216 **D**

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(38) For expansion $(x - \frac{3}{x})^8$

$$T_{r+1} = {}^8C_r (x)^{8-r} \left(-\frac{3}{x}\right)^r$$

$$= {}^8C_r \, x^{8-r} \cdot x^{-r} (-3)^r$$

$$= {}^8C_r \, x^{8-2r} (-3)^r$$

In order to get an independent term of x,

$x^{8-2r} = 1 = x^0$

then $8 - 2r = 0 \qquad r = 4$

Hence, the term independent of x is

$$T_{4+1} = {}^8C_4 \, x^{8-2(4)} (-3)^4$$

$$= (70)\, x^0 \, (81)$$

$$= 5670$$

B

(39) For expansion $(2x - 3y)^5$

$$T_{r+1} = {}^5C_r (2x)^{5-r} (-3y)^r$$

In order to have the coefficient of x^3y^2 (where power of $y = 2$), $r = 2$

Hence

$$T_{2+1} = {}^5C_2 (2x)^{5-2} (-3y)^2$$

$$= (10)(8x^3)(9y^2)$$

$$= 720x^3y^2$$

The coefficient of x^3y^2 is 720 C

(40) $(1+x)^{1/3} = 1 + \frac{1/3}{1}x + \frac{(1/3)(1/3-1)}{2}x^2$

$$= 1 + \frac{1}{3}x - \frac{1}{9}x^2$$

The coefficient of x^2 is $-\frac{1}{9}$ C

NOTE:

Given the two roots of a quadratic equation, let say α and β, the equation can be derived using

$x^2 - (\text{sum of roots})x + (\text{product of roots}) = 0$

i.e

$x^2 - (\alpha + \beta)x + \alpha\beta = 0$

(41) $2x^2 + 7x + 3 = 0$

Dividing by coefficient of x^2 (which is 2)

$$x^2 + \frac{7}{2}x + \frac{3}{2} = 0$$

Hence: $\alpha + \beta = -\frac{7}{2}$ and $\alpha\beta = \frac{3}{2}$

For the new equation:

the roots are $(2\alpha + \beta)$ and $(2\beta + \alpha)$

Product of roots:

$(2\alpha + \beta)(2\beta + \alpha)$

$= 4\alpha\beta + 2\alpha^2 + 2\beta^2 + \alpha\beta$

$= 5\alpha\beta + 2(\alpha^2 + \beta^2)$

[Recall: $\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta$]

$= 5\alpha\beta + 2[(\alpha+\beta)^2 - 2\alpha\beta]$

$= 5\alpha\beta + 2(\alpha+\beta)^2 - 4\alpha\beta$

$= 2(\alpha+\beta)^2 + \alpha\beta$

$= 2\left(-\frac{7}{2}\right)^2 + \frac{3}{2} = \frac{49}{2} + \frac{3}{2}$

$= \frac{52}{2} = 26$

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Sum of roots:

$(2\alpha+\beta)+(2\beta+\alpha) = 3\alpha+3\beta$

$= 3(\alpha+\beta) = 3\left(\frac{-7}{2}\right) = \frac{-21}{2}$

The new equation becomes

$x^2 - (\text{sum of roots})x + (\text{product of roots}) = 0$

$x^2 - \left(\frac{-21}{2}\right)x + 26 = 0$

$2x^2 + 21x + 52 = 0$ **D**

42. $ax^2 + (a+1)x + a = 0$

Dividing through by coefficient of x^2 $(=a)$

$x^2 + \left(\frac{a+1}{a}\right)x + 1 = 0$

Let α be the roots of the equation (since the two roots are equal)

Product of roots: $\alpha^2 = 1$

$\alpha = \pm\sqrt{1}$ $\alpha = +1$ or -1

Sum of roots:

$2\alpha = \frac{a+1}{a}$

when $\alpha = +1$

$2(1) = \frac{a+1}{a}$

$2a = a+1$

$2a - a = 1$

$a = 1$

when $\alpha = -1$

$2(-1) = \frac{a+1}{a}$

$-2a = a+1$

$-3a = 1$

$a = -1/3$

$a = -1/3$ or 1 **C**

43. $2x^2 - 6x + k = 0$

Dividing through by the coefficient of x^2 $(=2)$

$x^2 - 3x + k/2 = 0$

Product of roots: $\alpha\beta = k/2$ —①

43. Sum of roots: $\alpha+\beta = 3$ —(II)

Also given is $\frac{1}{\alpha} + \frac{1}{\beta} = -3$ —(III)

from (III): $\frac{\alpha+\beta}{\alpha\beta} = -3$

$\alpha+\beta = -3(\alpha\beta)$ —(IV)

Putting (I) and (II) into (IV)

$3 = -3(k/2)$

$k = 2$ **C**

44. $4x^2 - 6x + 1 = 0$

Dividing through by coefficient of x^2 $(=4)$

$x^2 - \frac{6}{4}x + \frac{1}{4} = 0$

Product of roots: $\alpha\beta = 1/4$

Sum of roots: $\alpha+\beta = 6/4$

For the new equation

Roots given are $\left(\alpha+\frac{1}{\beta}\right)$ and $\left(\beta+\frac{1}{\alpha}\right)$

Product of roots:

$\left(\alpha+\frac{1}{\beta}\right)\left(\beta+\frac{1}{\alpha}\right) = \alpha\beta + 1 + 1 + \frac{1}{\alpha\beta}$

$= 4 + 1 + 1 + \frac{1}{4} = \frac{25}{4}$

Sum of roots:

$\left(\alpha+\frac{1}{\beta}\right)+\left(\beta+\frac{1}{\alpha}\right) = (\alpha+\beta)+\left(\frac{1}{\alpha}+\frac{1}{\beta}\right)$

$= (\alpha+\beta) + \frac{\alpha+\beta}{\alpha\beta} = \frac{6}{4} + \frac{6/4}{1/4}$

$= \frac{3}{2} + 6 = \frac{15}{2}$

Hence, the new equation required is

$x^2 - (\text{sum of roots})x + (\text{product of roots}) = 0$

$x^2 - \frac{15}{2}x + \frac{25}{4} = 0$

$4x^2 - 30x + 25 = 0$ **B**

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45) $\frac{x-5}{(x^2-4)(x^2-9)} = \frac{A}{x-2} + \frac{B}{x+2} + \frac{C}{x-3} + \frac{D}{x+3}$

Then

$x-5 = A(x+2)(x^2-9) + B(x-2)(x^2-9) + C(x^2-4)(x+3) + D(x^2-4)(x-3)$

when $x = 2$

$2-5 = A(2+2)(2^2-9) + 0 + 0 + 0$

$-3 = A(4)(-5)$ $\quad A = 3/20$

when $x = -2$

$-2-5 = 0 + B(-2-2)(-2^2-9) + 0 + 0$

$-7 = B(-4)(-5)$ $\quad B = -7/20$

Hence $(A, B) = (3/20, -7/20)$ A

46) $\frac{x^2}{(x+1)^3} = \frac{A}{(x+1)} + \frac{B}{(x+1)^2} + \frac{C}{(x+1)^3}$

$x^2 = A(x+1)^2 + B(x+1) + C$

$x^2 = A(x^2+2x+1) + B(x+1) + C$

~~Compar~~ $= Ax^2 + 2Ax + A + Bx + B + C$

$x^2 = Ax^2 + (2A+B)x + (A+B+C)$

comparing the coefficients of x^2:

$A = 1$

comparing the coefficients of x:

$2A + B = 0$

$2(1) + B = 0$ $\quad B = -2$

comparing the constant terms:

$A + B + C = 0$

$1 + (-2) + C = 0$ $\quad C = 1$

Hence $(B, C) = (-2, 1)$

A

47) $\frac{3x^2-3x-2}{(x-1)(x+1)} = A + \frac{Bx+C}{(x-1)(x+1)}$

Using long division method:

$(x-1)(x+1) = x^2 - 1$

$$\begin{array}{r} 3 \\ x^2-1 \overline{)\,3x^2-3x-2} \\ -(3x^2 \quad -3) \\ \hline -3x+1 \end{array}$$

Then

$\frac{3x^2-3x-2}{(x-1)(x+1)} = 3 + \frac{-3x+1}{(x-1)(x+1)}$

$= A + \frac{Bx+C}{(x-1)(x+1)}$

Hence:

$A = 3, \quad B = -3, \quad C = 1$

D

48) $\frac{6x-10}{x^2-2x-3} = \frac{A}{x-3} + \frac{B}{x+1}$

$6x-10 = A(x+1) + B(x-3)$

when $x = 3$:

$6(3)-10 = A(3+1) + 0$

$18-10 = 4A, \quad A = 2$

when $x = -1$:

$6(-1)-10 = 0 + B(-1-3)$

$-6-10 = -4B, \quad B = 4$

Hence

$A = 2, \quad B = 4$

C

49) $\frac{7-2x}{(x+1)(x-2)^2} = \frac{A}{x+1} + \frac{B}{x-2} + \frac{C}{(x-2)^2}$

$7-2x = A(x-2)^2 + B(x+1)(x-2) + C(x+1)$

when $x = 2$:

$7-2(2) = 0 + 0 + C(2+1)$

$7-4 = 3C, \quad C = 1$

when $x = -1$

$7-2(-1) = A(-1-2)^2 + 0 + 0$

$7+2 = A(-3)^2 = 9A, \quad A = 1$

comparing the coefficient of x^2:

$A + B = 0$

$1 + B = 0 \quad B = -1$

Hence $A = 1, B = -1, C = 1$ D

50) $\frac{1}{(ax+b)^2} = \frac{A}{(ax+b)} + \frac{C}{(ax+b)^2}$

D

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